



INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR
Mid-Spring Semester 2025-26

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Date of Examination: 20-02-2026

Session(FN/AN) FN, Duration 2 hrs, Marks = 40

Subject Name : Privacy Enabling Technologies

Subject No : CS60212

Department/Centre/School : Computer Science and Engineering

Specific charts, graph paper, log book etc. required NO

Special Instructions (if any) In case of reasonable doubt, make assumptions and state them upfront.
Marks will be deducted for sketchy proofs and claims without proper reasoning.

1. Answer the following questions briefly and to the point:

- In terms of a given lattice L of dimension n and a target point t (not necessarily a lattice point), state the formulation of the Closest Vector Problem (CVP) [1]
- State (in words) the *Decisional* version of the LWE problem. Give equations (using a, b, e as appropriate) to denote the two sample-families which are formalized to be distinguished under the Decisional LWE problem. [1+1]
- Give equations to argue that Paillier cryptosystem exhibits additive homomorphism. [1]
- For TLWE encryption, give the equations for encoding and decoding elements from T_p to torus T_q where p divides q . [1]
- For some $p \in \mathbb{Z}$, define a plaintext space as $P = \mathbb{Z}/p\mathbb{Z}$. Then for some $m_1, m_2 \in P$, let $c_1 = TGSW(m_1)$ and $c_2 = TGSW(m_2)$, where $TGSW(\cdot)$ refers to the generic TGSW encryption as discussed in class. Let $c_3 = G^{-1}(c_2) \cdot c_1$. Prove that c_3 is the TGSW encryption of $m_1 + m_2 \pmod{p}$. Here G^{-1} is the inverse transformation applicable on gadget decomposition, as discussed in class. For full credit, give a general proof. [2]
- Given two TGLWE ciphertexts c_1 and c_2 and a TGGSW ciphertext c_b (of a single bit $\{0, 1\}$), provide the equations to derive CMUX operation to select one of $\{c_1, c_2\}$ based on c_b . Clearly mark and explain the operations involving the different ciphertexts. [2]
- State the equations for Gentry's decryption. [1]

2. You are given description of the following LWE based cryptosystem:

- Keygen:** Sample a secret vector $s \in \mathbb{Z}_q^n$ and public vector $a \in \mathbb{Z}_q^n$. Sample a secret error scalar $e \in \mathcal{X}$. Compute secret public scalar b as $b = \langle a, s \rangle + e \pmod{q}$, where q is the (publicly known) modulus.
- Encrypt:** To encrypt a single-bit message $m \in \{0, 1\}$, compute the ciphertext as $c = b + m \cdot \lfloor \frac{q}{2} \rfloor \pmod{q}$.

$$b = \langle a, s \rangle + m \cdot e$$

Given this context, answer the following:

- Give the equation for **Decrypt** function and give the necessary condition on e for correctness. [1+1]
- After applying your decryption equation as in (a), state the center of the distributions for **both** $m = 0$ and $m = 1$ (i.e. at what point in $\mathbb{Z}_q$ are the *noisy* distributions for $m = 0$ and $m = 1$ centered respectively). Assume that $\mathcal{X}$ is sampled from $\{-1, 0, 1\}$. [1+1]
- Modify the **Encrypt** equation above so that $m = 0$ is centered at $\frac{q}{4}$ and $m = 1$ is centered at $\frac{3q}{4}$ respectively. Likewise, modify the decoding ranges in $\mathbb{Z}_q$ for $m = 0$ and $m = 1$ in the **Decrypt** function in (a) to ensure correct decryption (do **not** modify the decryption equation; only modify the error range in $\mathbb{Z}_q$ within which you would decode to $m = 0$ or $m = 1$). [1+1]

$$1 \frac{q}{4}$$

$$c = b \pmod{q}$$

$$c = (b + \lfloor \frac{q}{2} \rfloor) \pmod{q}$$

$$\frac{q}{4} + \frac{q}{2}$$



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- Encrypt:** To encrypt a single-bit message $m \in \{0, 1\}$, compute the ciphertext as $c = b + m \cdot \lfloor \frac{q}{2} \rfloor \pmod{q}$

$$b = \langle a, s \rangle + m \Delta + e$$

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$$1 \frac{q}{4} \leftarrow c \bmod q$$

$$c = (b + \lfloor \frac{q}{2} \rfloor) \bmod q$$

$$\frac{q}{4} + \frac{q}{2}$$